

Vinay Kumar R

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EDUCATION

Stony Brook University

Master of Science in Data Science GPA: 3.6/4

Coursework: Algorithms, Operating Systems, Distributed Systems, Software Engineering

Stony Brook, NY

May 2025

Vellore Institute of Technology

Bachelor of Technology in Computer Science GPA: 9.1/10

Coursework: Data Structures, Database Management Systems, Object Oriented Design, Networks

Vellore, India

July 2023

EXPERIENCE

AI Innovation Institute

Research Assistant

Stony Brook, NY

September 2024 - Present

- Designed and deployed high-performance, fault-tolerant inference systems using GKE and Linux containers, optimizing CPU usage and system throughput for low-latency execution
- Developed CI/CD workflows with automated integration and system-level testing, improving developer velocity and deployment safety.
- Engineered TensorFlow parsers and 3D quaternion-to-trajectory converters, improving predictive analytics visualization for robotic motion.

YiPi Bear

Software Engineer

Remote

May 2024 - Present

- Spearheaded development of an AI-driven front-end and back-end platform using React, FastAPI, and MongoDB to manage end-to-end user interactions in a learning-focused environment
- Developed and deployed containerized Linux-based services using Docker and Kubernetes, with system-level tuning for performance and scalability.
- Wrote integration test plans, monitored system behavior under load, and documented design decisions to ensure safe and scalable deployments.

Dorklab

SDE Intern

Remote

December 2021 - February 2022

- Enhanced company's website by implementing critical features and resolving bugs on SpringBoot based microservices using Jenkins as CI/CD pipeline, PostgreSQL as database, Kafka for queue and event handling, significantly improving performance and user experience.

SKILLS

Programming Languages: Python, Java, C++, JavaScript, R, SQL, Scala, System Programming (C/C++), HTML, CSS

Web Technologies: HTTP, AJAX, DOM Manipulation, Web Services, JWT Authentication

Tools & Frameworks: AWS, GCP, Docker, Kubernetes, Flask, FastAPI, Django, React, Angular, MongoDB, PostgreSQL, CI/CD (GitHub Actions, Jenkins), Kafka, Elasticsearch, Linux, Oracle.

Concepts: Reinforcement Learning, LLMs, World Modeling, Distributed Systems, Software Engineering, Agile Development.

RELEVANT PROJECTS

Interactive Storytelling Platform

- Developed a scalable, full-stack storytelling platform integrating LLAMA and Waifu Diffusion through containerized microservices, enhancing interactive narrative learning experiences.
- Reduced response latency by 40% via prompt caching and asynchronous processing, enabling real-time narrative generation.
- Designed genre-specific prompt-chaining algorithms to generate branching stories, optimizing user experience

ZoneClock

- Engineered a Chrome extension that dynamically displays real-time local timestamps next to WhatsApp Web contacts, improving cross-timezone communication for global users.
- Leveraged JavaScript and phone-number-to-timezone API to dynamically convert phone numbers to geo-localized timestamps with 95% accuracy, included unit tests for timezone conversion modules.
- Optimized UI responsiveness by implementing MutationObservers and Chrome APIs to monitor and update DOM changes, improving update speed by 30% in dynamic content environments.

Moviepedia

- Developed front-end interfaces with React.js, incorporating JWT authentication to personalize content based on user watch history. Engineered database architecture and implemented CRUD operations for 5 entities using Node.js.

aRAGorn

- Created a chatbot using GPT-3.5 and BM25 for contextual argument generation from large structured data (Kialo).
- Built prompt chaining with few-shot learning and chain-of-thought reasoning to tackle ambiguous input scenarios effectively.

Careers Website

- Created a job portal with location-based search and one-click apply, integrating Google Maps API and Flask-based backend.
- Reduced average page load time by 35% via frontend optimizations and backend query tuning.
- Integrated Google location services to offer nearby job suggestions and implemented profile-based email notifications for relevant job opportunities, enabling one-click applications.